MIL-D-51468B(EA)
22 September 1986
SUPERSEDING
MIL-D-51468A(EA)
4 June 1986

MILITARY SPECIFICATION

DECONTAMINATION KIT, PERSONAL, M258A1

This specification is approved for use within Chemical Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of kit used for decontamination of chemical agents on skin and personal equipment.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Standards. The following standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1168 - Ammunition Lot Numbering

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, U.S. Army Chemical Research, Development and Engineering Center, ATTN: SMCCR-SPD-TS, Aberdeen Proving Ground, MD 21010-5423 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter. :

AMSC N/A FSC 4230

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2.1.2 Technical data package lists and publications. The following technical data package lists and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

TECHNICAL DATA PACKAGE LISTS (TDPL)

U.S. ARMY ARMAMENT, MUNITIONS AND CHEMICAL COMMAND

CHEMICAL RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

TDPL 5-77-2366 - Decontamination Kit, Personal, M258A1

PUBLICATIONS

U.S. ARMY ARMAMENT, MUNITIONS AND CHEMICAL COMMAND

CHEMICAL RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

SPECIAL PACKAGING INSTRUCTIONS (SPI)

P5-77-2366 - Decontamination Kit, Personal, M258A1

(Copies of standards, technical data package lists, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on TDPL 5-77-2366 or on the subsidiary drawings shall conform to the specifications listed thereon or to the specific characteristics set forth on the drawings.

3.1.2 Components. All components of the decontamination kit shall conform to the specifications and drawings listed on TDPL 5-77-2366 and subsidiary drawings.

3.2 Manufacture and assembly. The kit shall be manufactured and assembled in accordance with Drawing 5-77-2366 and subsidiary drawings.

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3.3 Ampoule leakage. The ampoules shall not leak when tested as specified in 4.4.4.1.

3.4 Ampoule fill. Each ampoule shall be filled with decontaminating solution to a height of 43 millimeters (mm) to 51 mm, when tested as specified in 4.4.4.2.

3.5 Decon 1 packet leakage. The decon 1 packet shall not leak when tested as specified in 4.4.4.3.

3.6 Decon 2 packet leakage. The decon 2 packet shall not leak when tested as specified in 4.4.4.3.

3.7 Decon 1 packet decontaminating solution content. The pad in the decon 1 packet shall contain 3.5 to 4.9 grams (g) of decontaminating solution when tested as specified in 4.4.4.4.

3.8 Decon 2 packet active chlorine content. The pad in the decon 2 packet shall have an active chlorine content of no less than 0.156 g when tested as specified in 4.4.4.5.

3.9 Strap tension. The welded area of the strap shall withstand a dead load of 18.0 $\pm$ 1.0 kilograms (kg) for 5 $\pm$ 1 seconds when tested as specified in 4.4.4.6.

3.10 First packet withdrawal. No other packets shall be pulled completely out of the kit when the first decon 1 packet is withdrawn from a filled kit when tested as specified in 4.4.4.7.

3.11 Workmanship. The decontamination kit and all components and subassemblies thereof shall be free from defects and contamination by foreign matter such as dirt, grease, and oil.

3.12 Preproduction. Prior to the start of regular production, a preproduction sample of kits shall be produced in accordance with this specification for examination and test (see 4.3).

3.12.1 Accelerated aging. The decontamination kit shall meet all the test requirements in this specification after accelerated aging as specified in 4.3.3.1.

3.12.2 Cold storage stability. The decontamination kit shall meet all the test requirements in this specification after cold storage as specified in 4.3.3.2.

3.13 Pull test. The assembled kit shall require 12 $\pm$ 6 pounds to open when tested in accordance with 4.4.4.8.

3.14 Pad sticking. During removal, the pad (towelette) shall not stick to the packet when tested as specified in 4.4.4.9.

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3.15 Ampoule tip. The sealed end of the ampoule shall be rounded and smooth to the touch when tested as specified in 4.4.4.10.

3.16 Ampoule screen. The screen shall show no evidence of damage or separation in the heat seal area when tested as specified in 4.4.4.11.

3.17 Packet components. There shall be no missing or damaged decon I and decon II packet components when tested as specified in 4.4.4.12.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3)
(b) Quality conformance inspection (see 4.4)

4.3 Preproduction inspection.

4.3.1 Lotting. A preproduction lot of no less than 1,000 kits shall be produced using the same methods, materials, equipment, and processes of manufacture as will be used during regular production.

4.3.2 Sampling and inspection procedure.

4.3.2.1 For examination. Sampling and inspection procedure for examination shall be as specified in 4.4.2.1 and 4.4.3.1.

4.3.2.2 For test. Sampling and inspection procedure for test shall be as specified in table I.

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TABLE I. Preproduction inspection

Requirement	Sample size	Accept/reject criteria	Test paragraph
Accelerated aging (3.12.1)	10 complete kits		4.3.3.1
Cold storage stability (3.12.2)	10 complete kits		4.3.3.2
Tests after accelerated aging or cold storage:			
Ampoule leakage (3.3)	90	Acc 2, rej 3	4.4.4.1
Ampoule fill (3.4)	30	Acc 1, rej 2	4.4.4.2
Decon 1 packet leakage (3.5)	30	Acc 1, rej 2	4.4.4.3
Decon 2 packet leakage (3.6)	30	Acc 1, rej 2	4.4.4.3
Decon 1 packet decontaminating solution content (3.7)	10 (1 per kit)	Acc 0, rej 1	4.4.4.4
Decon 2 packet active chlorine content (3.8)	10 (1 per kit)	Acc 0, rej 1	4.4.4.5
Strap tension (3.9)	10	Acc 0, rej 1	4.4.4.6
First packet withdrawal (3.10)	10	Acc 0, rej 1	4.4.4.7
Pull test (3.13)	10	Acc 0, rej 1	4.4.4.8
Pad sticking (3.14)	30* Decon I packets		
	30** Decon II packets	Acc 0, rej 1	4.4.4.9
Ampoule tip (3.15)	90***	Acc 0, rej 1	4.4.4.10
Ampoule screen (3.16)	30** Decon II packets	Acc 1, rej 2	4.4.4.11
Packet components (3.17)	30* Decon I packets		
	30** Decon II packets	Acc 0, rej 1	4.4.4.12

*Same sample packets used for decon I leakage test may be used.
**Same sample packets used for decon II leakage test may be used.
***Same sample packets used for ampoule leakage test may be used.

4.3.3 Tests. Tests shall be conducted as follows:

4.3.3.1 Accelerated aging. Place the decontamination kits in an oven at 43° ± 1°C for 24 hours. Remove the kits from the oven and allow to reach ambient temperature.

4.3.3.2 Cold storage stability. Store the decontamination kits at -54° ± 1°C for 24 hours. Remove the kits from cold storage and allow to reach ambient temperature.

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4.4 Quality conformance inspection.

4.4.1 Lotting. A lot of kits or components shall consist of the kits or the components produced by one manufacturer, at one plant, under essentially the same manufacturing conditions. However, no more than one interfix number of decon I packets or decon II packets shall be represented in any one lot of kits. Each lot shall be identified and controlled in accordance with MIL-STD-1168.

4.4.2 Sampling.

4.4.2.1 For examination. Sampling shall be conducted in accordance with MIL-STD-105.

4.4.2.2 For test. See 6.4 for sampling and testing precautions. Sampling shall be conducted in accordance with table II.

4.4.3 Inspection procedure.

4.4.3.1 For examination. Sample kits and components shall be examined in accordance with MIL-STD-105 and the following classification of defects. Defects 101 through 105, inclusive, may be examined prior to kit assembly.

(a) Decontamination kit, personal, M258A1 (Drawing 5-77-2366 and subsidiary drawings).

Categories		Defects
Critical:		None defined
Major:	AQL	1.0 percent defective
101		Packet heat seal width incorrect
102		Packet notch dimensions incorrect
103		Packet notch locations incorrect
104		Packet notches not cleanly cut
105		Packet marking incorrect, illegible, or missing
106		Damage to case or components missing
107		Lot numbering incorrect, illegible, or missing
108		Case marking incorrect, illegible, or missing
109		Kit components missing or incorrectly assembled
Minor:	AQL	2.5 percent defective
201		Components misplaced inside kit
202		Loose threads on cut edges of strap
203		Weld dimensions on strap incorrect
204		Foreign matter in or on case (see 3.11)

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TABLE II. Quality conformance inspection

Requirements	Sample size	Accept/reject criteria	Test paragraph
Ampoule leakage (3.3)	MIL-STD-105, prior to assembly in packets	AQL 1.0 percent defective	4.4.4.1
Ampoule fill (3.4)	MIL-STD-105, inspection level S-4, prior to assembly in packets	AQL 1.0 percent defective	4.4.4.2
Decon 1 packet leakage (3.5)	MIL-STD-105, inspection level S-4, prior to assembly in kits	AQL 2.5 percent defective	4.4.4.3
Decon 2 packet leakage (3.6)	MIL-STD-105, inspection level S-4, prior to assembly in kits	AQL 2.5 percent defective	4.4.4.3
Decon 1 packet decontaminating solution content (3.7)	20, prior to assembly in kits	Acc 1, rej 2	4.4.4.4
Decon 2 packet active chlorine content (3.8)	20, prior to assembly in kits	Acc 1, rej 2	4.4.4.5
Strap tension (3.9)	MIL-STD-105, inspection level S-3	AQL 2.5 percent defective	4.4.4.6
First packet withdrawal (3.10)	MIL-STD-105, inspection level S-4	AQL 2.5 percent defective	4.4.4.7
Pull test (3.13)	MIL-STD-105, inspection level S-4	AQL 1.5 percent defective	4.4.4.8
Pad sticking (3.14)			
Decon I packet	MIL-STD-105, inspection level S-4, prior to assembly in kits*	AQL 2.5 percent defective	4.4.4.9
Decon II packet	MIL-STD-105, inspection level S-4, prior to assembly in kits**	AQL 2.5 percent defective	4.4.4.9
Ampoule tip (3.15)	MIL-STD-105, prior to assembly in packets***	AQL 2.5 percent defective	4.4.4.10
Ampoule screen (3.16)	MIL-STD-105, inspection level S-4, prior to assembly in kits**	AQL 2.5 percent defective	4.4.4.11

*Same sample packets used for decon I leakage test may be used.
**Same sample packets used for decon II leakage test may be used.
***Same sample packets used for ampoule leakage test may be used.

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TABLE II. Quality conformance inspection - Continued

Requirements	Sample size	Accept/reject criteria	Test paragraph
Packet components (3.17)			
Decon I packet	MIL-STD-105, inspection level S-4, prior to assembly in kits*	AQL 2.5 percent defective	4.4.4.12
Decon II packet	MIL-STD-105, inspection level S-4, prior to assembly in kits**	AQL 2.5 percent defective	4.4.4.12

*Same sample packets used for decon I leakage test may be used.
**Same sample packets used for decon II leakage test may be used.

(b) Packaging inspection (section 5).

Categories	Defects
Critical:	None defined
Major:	AQL 4.0 percent defective
101	Unit packing or packing container not as specified
102	Contents of unit packing or packing container not as specified
103	Unit packing or packing container closure not as specified
104	Unit packing or packing container damaged
105	Marking missing, incorrect, or illegible
106	Fiberboard packing aids missing (where required) or not as specified
107	Shelf life marking missing, incorrect or illegible

4.4.3.2 For test. Sample kits shall be tested as specified in table II.

4.4.4 Tests. See 6.4 for sampling and testing precautions. Tests shall be conducted as follows:

4.4.4.1 Ampoule leakage. Place the ampoules in a vacuum tank and weight with a suitable cover to ensure complete submersion when the tank is filled with water. Place sufficient water soluble dye in the water to impart visible color to a drop of solution. Reduce the pressure in the tank to 127 mm of mercury or less absolute, maintain at that pressure for at least 25 minutes, and then pressurize to ambient. After no less than 20 minutes at ambient pressure, remove the ampoules from the tank, rinse thoroughly in clean water, and spread on a drying screen. Colored solution in an ampoule shall be considered evidence of leakage.

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4.4.4.2 Ampoule fill. With the ampoule held vertically on a flat surface, measure the height of the meniscus above the surface to the nearest millimeter.

4.4.4.3 Packet leakage. Submerge the packets for at least 30 seconds in ambient temperature water in a transparent vacuum tank under a vacuum of 7 ± 1 pounds per square inch. Keep the packet submerged in a vertical position with the uppermost surface covered by no more than 25 mm of water. Continuously examine the packet during the time it is submerged for evidence of bubbles. The presence of bubbles shall be considered evidence of leakage.

4.4.4.4 Decon 1 packet decontaminating solution content. Weigh the packet to the nearest 0.01 g. Open the packet and remove the pad. Wash the pad thoroughly with water. Place the pad and opened packet in an air circulation oven and dry at 105° ± 2°C for 60 ± 5 minutes. Weigh the entire packet and pad to the nearest 0.01 g. Calculate the weight of the solution on the pad as follows:

$$\text{Weight of solution} = 1.07(A - B)$$

where: A = Weight of pad and packet before drying, in grams, and
B = Weight of pad and packet after washing and drying, in grams.

4.4.4.5 Decon 2 packet active chlorine content. Remove the pad from the packet and place in a 250-milliliter (mL) flask. Add about 50 mL of water to the flask and mix thoroughly. Add 20 mL of a 10-percent aqueous solution of potassium iodide and 3 mL of glacial acetic acid, and allow to stand for 2 to 5 minutes. Titrate the liberated iodine with 0.1N sodium thiosulfate solution using starch indicator. Calculate the active chlorine content as follows:

$$\text{Active chlorine content in grams} = 0.0176AB$$

where: A = Milliliters of sodium thiosulfate solution used in titration, and
B = Normality of the sodium thiosulfate solution.

4.4.4.6 Strap tension. Cut the strap off the kit so that the welded area is approximately in the center of the strap sample. Attach each end of the strap to a suitable device and apply a load of 18.0 ± 1.0 kg for 5 ± 1 seconds. The test may be performed without cutting the strap from the kit provided that the specified load is applied to the welded area.

4.4.4.7 First packet withdrawal. Hold the kit in an upright position and withdraw the first decon 1 packet nearest the front of the kit. Observe if any of the remaining packets have been pulled completely out of the kit.

4.4.4.8 Pull test. The cover shall be pulled from the case at a rate of 12 ± 2 inches/minute. The suitability of the pull tester and fixture for holding the case and cover shall be determined by the responsible Quality Assurance Office. Operating conditions shall be 70 ± 20°F and 50 ± 20 percent relative humidity. A lubricant shall not be used on the parts to facilitate the pull test.

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4.4.4.9 Pad sticking. Open the packet and remove the pad (towelette).

4.4.4.10 Ampoule tip. Hold the ampoule and touch the sealed end. Inspect for evidence of a pointed tip.

4.4.4.11 Ampoule screen. Hold the screen and observe for compliance to the specified requirement.

4.4.4.12 Packet components. Empty the packets and visually inspect the components.

5. PACKAGING

5.1 Packaging. Packaging shall be as specified in SPI P5-77-2366.

6. NOTES

6.1 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Level of preservation and packing required (see 5.1).
(c) Shelf life code.
(d) Palletization.

6.2 Component specifications.

DOD-C-51464 - Chloramine B (Metric)
DOD-D-51467 - Decontaminating Solutions (Metric)

6.3 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value should be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of ASTM E 29.

6.4 Sampling and testing precautions. This specification covers inspection of chemical material which is potentially hazardous to personnel (toxic by ingestion, eye irritant). All applicable safety rules, regulations, and procedures must be followed in the handling and processing of this material. Refer to appropriate material safety data sheets for any safety precautions.

6.5 Subject term (key word) listing.

Decontamination
Decontamination kits

Custodian:

Army - EA

Preparing activity:

Army - EA

Project No. 4230-A067

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-D-51468B(EA)	DECONTAMINATION KIT, PERSONAL, M258A1

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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